

Python Programming

Unit 06 – Lecture 04: Missing Data, Aggregation, Combining, CSV I/O

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Repository: <https://github.com/tali7c/Python-Programming>

Quick Links

Core Concepts

Demo

Interactive

Summary

Agenda

1 Core Concepts

2 Demo

3 Interactive

4 Summary

Learning Outcomes

- Detect and handle missing values (NaN)

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- Use aggregation (groupby) for summaries
- Combine DataFrames (concat, merge)
- Import/export CSV using `read_csv` and `to_csv`

Missing Data

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 - `dropna()`
 - `fillna(value)`

Aggregation with groupby

```
avg_by_city = df.groupby("city")["marks"].mean()
```

- Useful for summaries (average marks per city, counts per category)

Combining DataFrames

- `pd.concat([...])`: stack rows or columns

Combining DataFrames

- `pd.concat(...)`: stack rows or columns
- `pd.merge(a, b, on="key")`: SQL-like join

CSV Import/Export

```
df = pd.read_csv("data/student_scores.csv")  
df.to_csv("data/cleaned.csv", index=False)
```


Demo: Missing Data + Groupby + CSV

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 - reading CSV
 - missing value handling
 - groupby aggregation
 - merge with another DataFrame

Checkpoint 1

Question: What does `fillna(0)` do?

Checkpoint 2

Question: When would you use `merge` instead of `concat`?

Think-Pair-Share

Discuss:

- Is it always correct to replace missing marks with 0?
Why/why not?

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- `groupby` provides powerful summaries
- `merge` joins tables using keys; `concat` stacks data
- CSV I/O connects Pandas with real-world datasets

Exit Question

Write one line to read a CSV file named "marks.csv" into a DataFrame.